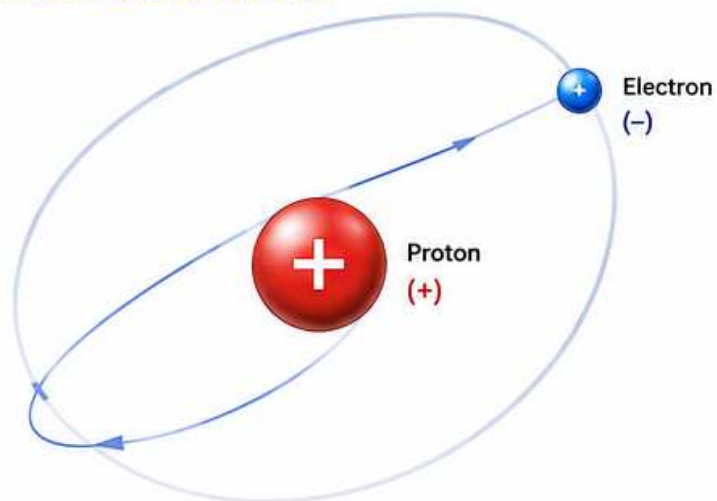


1.2 THE HYDROGEN ATOM

The building block of the MRI signal

MRI is all about the hydrogen nucleus (proton).
It has a positive charge and spins like a tiny magnet.
This spin creates a magnetic moment (μ).

THE HYDROGEN ATOM



MAGNETIC MOMENT (μ)

Because the proton spins, it behaves like a tiny magnet with a magnetic moment (μ).

$$\vec{\mu} = \gamma \vec{I}$$

Where:

γ = gyromagnetic ratio
(42.58 MHz/T for hydrogen)

$\vec{I}$ = nuclear spin



WHY HYDROGEN?

- ✓ Most abundant element in the body (water & fat)
- ✓ Single proton (simple and efficient)
- ✓ Strong magnetic moment (good signal)
- ✓ Safe and non-ionizing
- ✓ Present in almost all tissues

TYPICAL ABUNDANCE IN THE BODY



Water (H₂O)
~ 110,000,000,000,000
protons per mL



Fat (Lipids)
~ 60,000,000,000,000
protons per mL



Muscle (Tissue)
~ 70,000,000,000,000
protons per mL

KEY TAKEAWAY

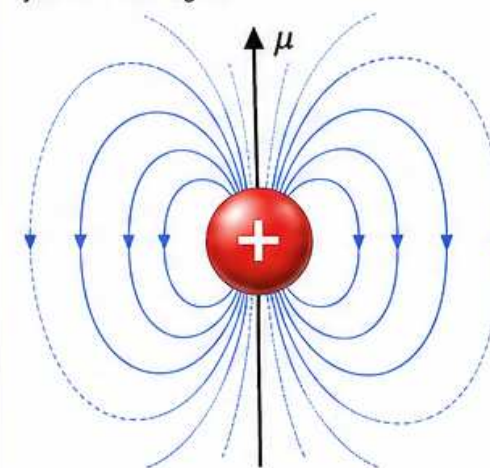
Every MRI signal we detect comes from the tiny spins of hydrogen protons as they interact with magnetic fields and radio waves.

HYDROGEN NUCLEUS (PROTON) PROPERTIES

Property	Value / Description
Charge	+1 (positive)
Spin	1/2
Magnetic Moment (μ)	5.05×10^{-27} J/T
Gyromagnetic Ratio (γ)	42.58 MHz/T
Natural Abundance	99.985% of all hydrogen nuclei
Resonant Frequency	63.87 MHz at 1.5 Tesla 127.74 MHz at 3.0 Tesla 298.06 MHz at 7.0 Tesla

PROTON AS A TINY BAR MAGNET

The spinning proton creates a magnetic field around itself, just like a bar magnet.



ENGINEERING INSIGHT

The strength of the MRI signal depends on how many protons are present and how well their magnetic moments can be controlled.



FIELD SERVICE TIP

The better the main magnetic field (B_0) uniformity, the better the protons align and the stronger the signal.



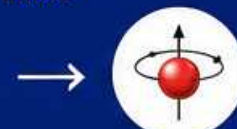
COMMON MISCONCEPTION

MRI does not "see" atoms. It detects the signals from the protons in hydrogen.

IN SIMPLE TERMS



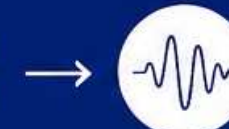
Hydrogen in the body



Proton spins like a magnet



Creates a magnetic moment



Interacts with B_0 and RF to produce MRI signal